

Part I

SUMMARY of Chain Rule

Summary

February 21, 2026

- **The Chain Rule:** If $f(x)$ and $g(x)$ are differentiable functions, then $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$.
- **The Derivatives of Trigonometric Functions:**

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

Part II

Recitation Questions

Problem 1

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Problem 1 For the following problems, the derivative is given. Determine which function was the original function.

(a) The derivative is $f'(x) = \cos(x)e^{\sin(x)}$. Which is the original function?

(i) $f(x) = (\sin(x))(e^x)$

(ii) $f(x) = \sin(e^x)$

(iii) $f(x) = e^{\sin(x)}$

(iv) $f(x) = e^{x \sin(x)}$

Explanation. Since

$$\frac{d}{dx} \left(e^{\sin(x)} \right) = e^{\sin(x)} \cdot \cos(x) = \cos(x)e^{\sin(x)}$$

the correct answer is (iii).

(b) The derivative is $g'(x) = 4 \left(\tan(x^4 - 5x) \right)^3 \sec^2(x^4 - 5x)(4x^3 - 5)$. Which is the original function?

(i) $g(x) = (\tan(x) - 5x)^4$

(ii) $g(x) = \tan^4(x) - 5x^4$

(iii) $g(x) = \tan(x^4 - 5x)$

(iv) $g(x) = \tan^4(x^4 - 5x)$

Explanation. Since

$$\frac{d}{dx} \left(\tan^4(x^4 - 5x) \right) = 4 \tan^3(x^4 - 5x) \sec^2(x^4 - 5x)(4x^3 - 5)$$

the correct answer is (iv). Notice that $\tan^3(x^4 - 5x) = (\tan(x^4 - 5x))^3$ are (slightly) different notations for the exact same expression.

Problem 2

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Problem 2 A table of values for $f(x)$ and $f'(x)$ is shown below:

x	$f(x)$	$f'(x)$
1	3	4
2	2	3
3	4	5
4	6	3

- Evaluate the limit $\lim_{x \rightarrow 2} \frac{f(x^2) - 6}{x - 2}$. **EXPLAIN.**

Explanation. If we set $g(x) = f(x^2)$, this limit is $\lim_{x \rightarrow 2} \frac{g(x) - g(2)}{x - 2}$. In other words, we're being asked to evaluate $\frac{d}{dx} f(x^2)$ at $x = 2$. By the Chain Rule, $\frac{d}{dx} f(x^2) = 2x f'(x^2)$. When evaluated at $x = 2$ this is $2(2)f'(4) = 4(3) = 12$. The value of this limit is 12.

Our full explanation is then: $\lim_{x \rightarrow 2} \frac{f(x^2)}{x - 2} = \left[\frac{d}{dx} f(x^2) \right]_{x=2}$ by the limit definition of the derivative. Our calculation above shows that this equals 12.

- Evaluate $\frac{d}{dx} f(f(x))$ at $x = 3$.
 - 6
 - 25
 - 5
 - 15
 - DNE
 - None of the previous answers.

Explanation. The answer is (d):

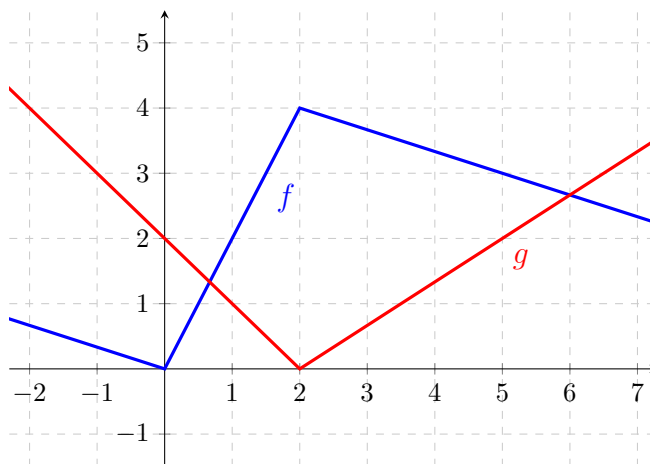
$$\begin{aligned}
 \left[\frac{d}{dx} f(f(x)) \right]_{x=3} &= f'(f(3)) \cdot f'(3) \\
 &= f'(4) \cdot 5 \\
 &= 3 \cdot 5 = 15
 \end{aligned}$$

Problem 3

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Problem 3

Problem 3 Given the following graphs of f and g (both piecewise linear functions), define new functions $u(x) = f(g(x))$ and $v(x) = f(x)g(x)$. Find:



(a) $u'(1)$

Explanation. $u'(x) = \frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$. So,

$$\begin{aligned} u'(1) &= f'(g(1)) \cdot g'(1) \\ &= f'(1) \cdot (-1) \\ &= (2)(-1) = -2 \end{aligned}$$

(b) $v'(1)$

Explanation. $v'(x) = \frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$. So,

$$\begin{aligned} v'(1) &= f'(1)g(1) + f(1)g'(1) \\ &= (2)(1) + (2)(-1) \\ &= 0 \end{aligned}$$

(c) $\lim_{x \rightarrow 1} \frac{\sqrt{g(x)} - 1}{x - 1}$

Explanation. This is asking for $\left[\frac{d}{dx}(\sqrt{g(x)}) \right]_{x=1}$. By the Chain Rule,

Problem 3

$$\frac{d}{dx}(\sqrt{g(x)}) = \frac{g'(x)}{2\sqrt{g(x)}} \text{ so}$$

$$\begin{aligned}\left[\frac{d}{dx}(\sqrt{g(x)})\right]_{x=1} &= \frac{g'(1)}{2\sqrt{g(1)}} \\ &= \frac{-1}{2\sqrt{1}} \\ &= -\frac{1}{2}.\end{aligned}$$

Problem 4

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Problem 4 Suppose the line tangent to the graph of f at $x = 1$ is $y = 6x - 7$. Find an equation of the line tangent to the following curves at $x = 1$:

(a) $y = g(x) = 5(f(x))^4$

Explanation. First, in the equation of the tangent line to $f(x)$ at $x = 1$, when $x = 1$ we have that $y = 6(1) - 7 = -1$. Thus $f(1) = -1$ and therefore $g(1) = 5(-1)^4 = 5$. Hence, a point on our line is $(1, 5)$. Also, since the slope of the given tangent line is $m = 6$, we know that $f'(1) = 6$.

Now, by the chain rule we have that $g'(x) = 5 \cdot 4(f(x))^3 \cdot f'(x)$. So $g'(1) = 20(f(1))^3 \cdot f'(1) = 20(-1)^3(6) = -120$. Thus, the equation of the line tangent to the graph of $y = g(x)$ at $x = 1$ is

$$y - 5 = -120(x - 1)$$

$$y = -120x + 125$$

(b) $y = h(x) = x^2(f(x^3))$

Explanation. $h(1) = 1^2 \cdot f(1^3) = f(1) = -1$. By the product and chain rules:

$$h'(x) = 2x(f(x^3)) + x^2(f'(x^3) \cdot 3x^2) = 2x(f(x^3)) + 3x^4(f'(x^3))$$

Thus, $h'(1) = 2f(1) + 3f'(1) = 2(-1) + 3(6) = -2 + 18 = 16$. So, the equation of the line tangent to the graph of h at $x = 1$ is

$$y - (-1) = 16(x - 1)$$

$$y = 16x - 17$$

Problem 5

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Problem 5 Differentiate each function (with respect to x)

(a) $\cos(\sqrt{x+7})$

Explanation.

$$\begin{aligned}\frac{d}{dx}(\cos(\sqrt{x+7})) &= -\sin(\sqrt{x+7}) \cdot \frac{1}{2}(x+7)^{-\frac{1}{2}}(1) \\ &= \frac{-\sin(\sqrt{x+7})}{2\sqrt{x+7}}\end{aligned}$$

(b) $\sqrt{\cos(x)+7}$

Explanation.

$$\begin{aligned}\frac{d}{dx}(\sqrt{\cos(x)+7}) &= \frac{1}{2}(\cos(x)+7)^{-\frac{1}{2}}(-\sin(x)) \\ &= \frac{-\sin(x)}{2\sqrt{\cos(x)+7}}\end{aligned}$$

(c) $\sqrt{\cos(x)}+7$

Explanation.

$$\begin{aligned}\frac{d}{dx}(\sqrt{\cos(x)}+7) &= \frac{1}{2}(\cos(x))^{-\frac{1}{2}}(-\sin(x)) + 0 \\ &= \frac{-\sin(x)}{2\sqrt{\cos(x)}}\end{aligned}$$

(d) $\cos(\sqrt{x}+7)$

Explanation.

$$\begin{aligned}\frac{d}{dx}(\cos(\sqrt{x}+7)) &= -\sin(\sqrt{x}+7) \cdot \frac{1}{2}x^{-\frac{1}{2}} \\ &= \frac{-\sin(\sqrt{x}+7)}{2\sqrt{x}}\end{aligned}$$

(e) $\cos(x) \cdot (\sqrt{x}+7)$

Explanation.

$$\begin{aligned}\frac{d}{dx}(\cos(x) \cdot (\sqrt{x}+7)) &= -\sin(x) \cdot (\sqrt{x}+7) + \cos(x) \cdot \frac{1}{2}x^{-\frac{1}{2}} \\ &= -\sin(x) \cdot (\sqrt{x}+7) + \frac{\cos(x)}{2\sqrt{x}}\end{aligned}$$

Problem 6

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Problem 6 Find the derivative of the following functions:

(a) $f(x) = \sin(x) \cos(x)$

Explanation.

$$f'(x) = (\cos(x))(\cos(x)) + (\sin(x))(-\sin(x)) = \cos^2(x) - \sin^2(x).$$

(b) $f(x) = \frac{e^x \tan(x)}{\sec(x) + 2}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{(\sec(x) + 2)(e^x \tan(x) + e^x \sec^2(x)) - e^x \tan(x)(\sec(x) \tan(x))}{(\sec(x) + 2)^2} \\ &= \frac{e^x[(\sec(x) + 2)(\tan(x) + \sec^2(x)) - \sec(x) \tan^2(x)]}{(\sec(x) + 2)^2}. \end{aligned}$$

(c) $f(x) = e^{x \tan(x)}$

Explanation.

$$f'(x) = e^{x \tan(x)} (\tan(x) + x \sec^2(x))$$

(d) $f(x) = \sin(x) \cos(x) e^{3x}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{d}{dx} [\sin(x) \cos(x)] e^{3x} + (\sin(x) \cos(x)) \frac{d}{dx} (e^{3x}) \\ &= (\cos^2(x) - \sin^2(x)) e^{3x} + 3e^{3x} \sin(x) \cos(x) \\ &= e^{3x} (\cos^2(x) + 3 \sin(x) \cos(x) - \sin^2(x)). \end{aligned}$$

(e) $f(x) = \frac{x + 5}{7x^6 + \cot(x)}$

Explanation.

$$\begin{aligned} f'(x) &= \frac{(7x^6 + \cot(x))(1) - (x + 5)(42x^5 - \csc^2(x))}{(7x^6 + \cot(x))^2} \\ &= \frac{7x^6 + \cot(x) - (x + 5)(42x^5 - \csc^2(x))}{(7x^6 + \cot(x))^2}. \end{aligned}$$

(f) $f(x) = \sin(2x) \sec^3(x^2 + 4x)$

Explanation.

$$\begin{aligned} f'(x) &= 2 \cos(2x) \sec^3(x^2 + 4x) + \sin(2x) 3 \sec^2(x^2 + 4x) \sec(x^2 + 4x) \tan(x^2 + 4x) (2x + 4) \\ &= 2 \cos(2x) \sec^3(x^2 + 4x) + 3 \sin(2x) \sec^3(x^2 + 4x) \tan(x^2 + 4x) (2x + 4). \end{aligned}$$

Problem 7

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Problem 7 Find values for a , b , and c so that the following function is differentiable everywhere.

$$f(x) = \begin{cases} a \sin(x) + b \cos(x) & \text{if } x < 0 \\ ax^2 + bx + c & \text{if } x \geq 0 \end{cases}$$

Explanation. The function f is differentiable for $x < 0$ since f is a combination of trigonometric functions. f is also differentiable $x > 0$ since f is a polynomial on that interval. We need to focus on $x = 0$. Since f is differentiable everywhere, f must also be differentiable at $x = 0$ and therefore continuous at $x = 0$. Therefore:

We need that $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} f(x) = f(0)$. Observe that

- $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (a \sin(x) + b \cos(x)) = b(1) = b.$
- $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (ax^2 + bx + c) = c.$

Thus, we must have that $b = c$

Next, if $f'(0)$ exists, then:

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$\text{since } f(0) = a \cdot 0^2 + b \cdot 0 + c = c$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - c}{x}$$

Since this limit exists, the left and right limits must be equal.

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{f(x) - c}{x} &= \lim_{x \rightarrow 0^+} \frac{ax^2 + bx + c - c}{x} \\ &= \lim_{x \rightarrow 0^+} \frac{ax^2 + bx}{x} \\ &= \lim_{x \rightarrow 0^+} (ax + b) \\ &= b \end{aligned}$$

Problem 7

$$\begin{aligned}\lim_{x \rightarrow 0^-} \frac{f(x) - c}{x} &= \lim_{x \rightarrow 0^-} \frac{a \sin(x) + b \cos(x) - c}{x} \\ &= \lim_{x \rightarrow 0^-} \frac{a \sin(x)}{x} + \lim_{x \rightarrow 0^-} \frac{b \cos(x) - c}{x}\end{aligned}$$

we already found $b = c$ so we have

$$\begin{aligned}\lim_{x \rightarrow 0^-} \frac{a \sin(x)}{x} + \lim_{x \rightarrow 0^-} \frac{c \cdot \cos(x) - c}{x} \\ = a \cdot \lim_{x \rightarrow 0^-} \frac{\sin(x)}{x} + c \cdot \lim_{x \rightarrow 0^-} \frac{\cos(x) - 1}{x}\end{aligned}$$

since $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x} = 0$ we have

$$a \cdot \lim_{x \rightarrow 0^-} \frac{\sin(x)}{x} + c \cdot \lim_{x \rightarrow 0^-} \frac{\cos(x) - 1}{x} = a$$

This means in order for the derivative to exist, $a = b = c$.